



Tackling missing data in PLS-SEM: strategies and insights for business research

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ABSTRACT

This study provides a practical guide for handling missing data in partial least squares structural equation modeling (PLS-SEM), a prominent multivariate technique that is widely used in business research. We compare the strengths and limitations of different missing data handling techniques, emphasizing the importance of selecting appropriate methods to enhance the accuracy and reliability of PLS-SEM analyses. Furthermore, we introduce an innovative approach for dealing with not missing at random (NMAR) data by combining imputation with subsequent weighting. By demonstrating the practical effects of various treatment strategies through empirical case studies and a comprehensive simulation study, this research offers meaningful insights and pragmatic guidelines for business researchers dealing with missing data in PLS-SEM.

1. Introduction

Handling missing data is a critical challenge in partial least squares structural equation modeling (PLS-SEM), a widely utilized multivariate statistical technique in business disciplines such as organizational behavior, marketing, human resource management, information systems, international business, tourism, and hospitality (Becker et al., 2023; Sarstedt & Liu, 2024). PLS-SEM is particularly valued for its ability to manage complex structural relationships or nomological net, making it indispensable for causal-predictive and intricate model analyses (Shmueli et al., 2019; Cheah et al., 2020; Adler et al., 2023; Lien-gaard, 2024; Sharma et al., 2024). However, the presence of missing data in survey-based research, a common practice in business studies, poses a significant threat to the precision and reliability of PLS-SEM results (Sarstedt et al., 2023).

Missing data in any study has the potential to introduce bias, reduce statistical power, and distort model estimations, which can consequently result in inaccurate or unreliable conclusions (Enders, 2023). The nature and extent of missingness—categorized as missing completely at random (MCAR), missing at random (MAR), or not missing at random (NMAR)—play a pivotal role in determining the appropriate handling

strategy (Fiero et al., 2016). Despite the growing use of various techniques, such as expectation maximization (EM) (Wang et al., 2022), multiple imputation by chained equations (MICE) (Grund et al., 2024), and others in conjunction with PLS methods for handling missing data in surveys, there remains a lack of comprehensive guidance specifically tailored to the needs of PLS-SEM users in business research (Sarstedt et al., 2023).

This study addresses the gap by providing an in-depth comparison of different missing data handling methods within the context of PLS-SEM. In addition, this study introduces an innovative approach for managing NMAR data by integrating advanced imputation techniques with subsequent weighting strategies, specifically through the EM-Weighting method. This dual-method approach, termed EM-Weighting, not only imputes the missing values based on underlying data patterns but also applies weighting to adjust for the biases introduced by the NMAR mechanism. By leveraging the advantages of weighted PLS-SEM (WPLS-SEM) as demonstrated by Becker and Ismail (2016) and Cheah et al. (2021), the efficacy of this approach is evidenced by the enhancement of the accuracy and reliability of parameter estimates. This provides researchers with practical guidelines to mitigate the adverse effects of non-random missingness in their analyses.

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Furthermore, the present research undertakes a comprehensive simulation study, the objective of which is to examine the effect of varying levels of missing data on commonly used imputation methods in PLS-SEM. To assess this approach, we conducted a comprehensive simulation study, using the NIPALS method as a benchmark to evaluate the performance of EM, MICE, and EM-Weighting across MCAR, MAR, and NMAR at 5 % to 50 % missingness levels. By systematically altering the proportion and mechanism of missing values, the simulation aims to identify the thresholds at which particular imputation techniques begin to yield unreliable or biased results or fail as the level of missing values increases. Ultimately, these findings can assist business researchers in making more informed decisions regarding the application of missing data techniques and the consideration of alternative strategies as the severity of missing data increases.

By employing both empirical case studies and simulations, this research demonstrates the practical implications of various data treatment strategies. This study not only validates the effectiveness of the proposed NMAR management approach but also contributes to the advancement of methodological practices in empirical business research, ensuring that analyses remain robust and dependable even in the presence of complex missing data patterns.

2. Types and identification of missing data

The effective management of missing data is of paramount importance for preserving the integrity of quantitative analyses. Such missing data can arise from non-response, response biases, data entry errors, or technical issues during data collection (Zhao, 2020). Incorrect handling of missing values can result in biased estimates, reduced statistical power, and compromised model validity (Enders, 2023). Therefore, it is imperative to accurately identify the mechanism behind missingness — whether it be MCAR, MAR, or NMAR— in order to select the most appropriate treatment methods.

2.1. Review of missing value reporting practices

To assess the current state of missing data handling in business research, this study reviews papers published in the *Journal of Business Research* from 2019 to 2023, focusing on those employing quantitative analysis methods. Of the 2,719 papers reviewed, 651 (24 %) reported data omissions in various forms. The distribution of missing data across different data sources and years is detailed in Tables 1 and 2.

Our key findings reveal that missing data remains a persistent issue in business research, with approximately 24 % of the reviewed papers reporting instances of data omission. This prevalence underscores the importance of addressing missing data to uphold the integrity of research outcomes. Additionally, the majority of studies utilize deletion methods, such as listwise deletion, to manage missing data, while advanced techniques like EM or multiple imputation are seldom employed. This predominant reliance on simplistic deletion methods highlights a critical need for the adoption of more sophisticated data handling approaches to enhance the quality and reliability of research findings.

These findings point to significant inconsistencies and a lack of transparency in current data handling practices within the field of business research. The dominance of deletion methods, coupled with the

Table 2
Percentage of missing data.

Percentage	2019	2020	2021	2022	2023
No Mention	8	16	19	49	26
<5%	10	15	10	16	10
5–10 %	50	29	39	43	54
10–20 %	38	41	10	38	40
20–40 %	0	0	9	0	0

underutilization of advanced imputation techniques, indicates a pressing need for future studies to implement and report more robust data-handling strategies. By improving these practices, researchers can not only bolster the validity of their results but also provide a more reliable foundation for empirical studies in the business domain. This enhancement of data handling methodologies will ultimately contribute to more accurate and trustworthy research outcomes, fostering greater confidence in empirical business studies.

2.2. Types of missing data

Missing data mechanisms are typically classified into three types—MCAR, MAR, and NMAR (Fiero et al., 2016). MCAR arises when missing data is randomly and independently distributed across both observed and unobserved variables (Austin, 2011; Davey & Savla, 2009). To illustrate, consider data entry errors resulting from survey respondents’ inadvertent input of information in a customer satisfaction survey. These errors are not systematically related to the content of the survey. While MCAR does not introduce bias in results, it does reduce statistical power due to a decrease in the sample size (Goretzko et al, 2020).

MAR occurs when missingness is related to observed but not unobserved data (Grund et al., 2025). Suppose in a product quality survey that older respondents or those who purchase more frequently are less likely to complete the questionnaire; if their tendency to drop out is unrelated to their (unobserved) satisfaction levels, then missingness is MAR. MAR data can often be handled using imputation strategies that model the relationship between missingness and observed variables, thus mitigating bias (Grund et al., 2025).

NMAR signifies that missing data is directly associated with unobserved information (Little & Rubin, 2019). To illustrate, dissatisfied customers may opt to omit responses to questions regarding “future visits” due to their dissatisfaction (unobserved) exerting influence on nonresponse. In such instances, the implementation of specialized approaches, such as weighting schemes or selection models, is imperative to address potential biases that standard imputation alone may not adequately address (Jansen & Molenberghs, 2010).

While MCAR and MAR mechanisms are often treated as *ignorable*, the NMAR mechanism demands *non-ignorable* methods because of the significant bias it can introduce (Little & Rubin, 2019). In the event of an inaccurate classification of the mechanism, there is a risk that inappropriate methods will be employed, resulting in unreliable or biased outcomes. It is important to note that while several testing methods exist—such as propensity score analysis and Little’s MCAR test—these tests should not be the sole criterion for determining the missing data mechanism (Little & Rubin, 2019; Jung et al., 2011). These methods

Table 1
Distribution of Data Sources.

Data sources	Number of papers reviewed					Number of papers mentioning missing values				
	2019	2020	2021	2022	2023	2019	2020	2021	2022	2023
Third-party	78	78	151	189	134	19 (24.36 %)	26(33.33 %)	28 (18.54 %)	41 (21.69 %)	40 (29.85 %)
Survey	233	294	346	291	155	76 (32.62 %)	93 (31.63 %)	58 (16.76 %)	48 (16.49 %)	52 (33.55 %)
Experiment	73	171	225	171	130	11 (15.07 %)	18 (10.53 %)	11 (4.89 %)	57 (33.33 %)	59 (45.38 %)
Total	384	543	722	651	419	106 (27.60 %)	137 (25.23 %)	97 (13.43 %)	146 (22.43 %)	151 (36.04 %)

cannot ensure precise identification of the missing data mechanism. In practice, researchers should view these tests as supplementary to theoretical considerations and a comprehensive understanding of the data collection process. Contextual knowledge regarding the reasons behind data missingness can offer critical insights, thereby enhancing the accuracy in identifying the underlying missing data mechanism.

Missing data poses a unique challenge for PLS-SEM, a method that is often used in prediction-focused studies involving complex models with numerous constructs, multiple paths, multiple indicators, multiple data groups or higher-order constructs (Hair et al., 2023). Even modest levels of missing data can propagate through latent variable scores, thereby undermining path estimates and the validity of the overall model. Some PLS-specific solutions use algorithms (e.g. NIPALS) to estimate latent scores iteratively. This reduces the need for listwise deletion (Wold, 1966). More approaches can integrate EM- or MICE-based imputation prior to PLS-SEM (Wang et al., 2022; Grund et al., 2025), which is particularly beneficial when the data are categorized as missing at random (MAR).

Another pivotal consideration in PLS-SEM pertains to the distinction between reflective and formative indicators (Hair et al., 2024). Given that each formative indicator is presumed to capture a distinct dimension or attribute of a construct, the absence of even a single indicator—or the occurrence of specific patterns of missingness—has the potential to fundamentally affect the measurement structure (Wang et al., 2022; Hair et al., 2024). The NIPALS algorithm is commonly used in PLS-SEM to iteratively estimate latent variable scores (e.g., for constructs such as *COMP*, *LIKE*, *CUSA*, and *CUSL*) by leveraging available data, even in the presence of missing values (Wold, 1966). Although NIPALS does not produce a fully imputed dataset at the indicator level (e.g., *comp_1*, *comp_2*, *comp_3*), the latent scores that are generated are robust and can be utilized for further predictive and exploratory analyses, such as ANOVA, MANOVA, profile analysis, or clustering. These latent scores often provide a more comprehensive representation of the construct than individual measurement variables, making NIPALS particularly valuable for business research applications with low missingness scenarios. The computation of these latent scores is tied to a basic fit measure, such as the variance explained by the latent variable for each indicator, which provides an early sign of how well the model fits the data and can guide further imputation or analysis steps. For example, if the variance explained for an indicator like firm competence (*COMP*) is low, it may suggest the indicator poorly reflects the concept, prompting researchers to refine the model—by adjusting indicator selection or enhancing imputation methods like weighting strategies to address missingness—or switch to a different approach before proceeding with further analyses. This is particularly relevant for reflective constructs, where latent scores can compensate for some missing data. However, for formative constructs, where each indicator's completeness is essential to define the construct accurately, formal imputation (e.g., via EM or MICE) may still be recommended to maintain the measurement structure's integrity.

In addition, the emphasis on prediction in PLS-SEM further accentuates the risk posed by missing data, particularly when key predictors or outcome variables are incomplete. Prediction techniques like PLSpredict (Shmueli et al., 2016; 2019) and cross-validated predictive ability test (CVPAT; Liengaard et al., 2021; Sharma et al., 2023) assist in assessing the extent to which predictive performance is diminished when data are missing. In cases of moderate missingness and MAR-based data, imputation can potentially restore acceptable predictive accuracy (Wang et al., 2022). However, in cases where nonresponse stems from unobserved factors (NMAR), more advanced or iterative methods—such as weighting or repeated re-estimation of latent constructs—may be necessary to preserve predictive power.

Challenges arising from software and implementation issues can compound these difficulties. Some PLS-SEM platforms default to listwise deletion, whereas others (particularly those using a NIPALS-based algorithm) allow for certain missing patterns without discarding entire

cases. Researchers can perform multiple imputation on datasets with missing values prior to constructing path models, using tools such as the *mice* package in R; an imputed dataset can then be generated for subsequent PLS-SEM analysis. Datasets with missing values, along with R code and detailed steps for performing the imputations, are included in the Appendix A. The Appendix A also contains the results of the PLS-SEM analysis process after imputation, offering transparency in how the missing data was handled and ensuring that the generated datasets can be easily replicated.

3. Methods for handling missing data

Approaches to handling missing data can be categorized into three distinct methods: deletion methods, imputation methods, and model-based approaches (Zhao, 2020). Each of these categories is characterized by its own unique set of assumptions, technical requirements, and consequences for the quality of the resulting analyses. The selection of the most appropriate method is contingent upon the nature of the missingness mechanism (MCAR, MAR, or NMAR), the proportion of missing data, and the complexity of the research model (Enders, 2023).

The deletion method is a straightforward process; however, it is not without risk. In the event that the missing values are not completely random, there is a possibility of information loss and biased results (Zhu et al., 2014). Imputation methods, on the other hand, replace missing values with estimates that are plausible, thereby preserving the sample size (Enders, 2023). However, this process requires careful modelling in order to avoid the introduction of artificial patterns. Model-based approaches, such as selection models or weighting techniques, explicitly model the missing data mechanism. They are particularly useful when dealing with NMAR data, where standard methods are unable to capture the underlying bias.

This section provides a discussion of the key methods relevant to PLS-SEM—including deletion, EM imputation, fully conditional specification (FCS)/MICE, and approaches tailored to NMAR data—along with practical considerations for their application. By understanding the strengths and limitations of each method, researchers can make informed decisions that enhance the reliability of their PLS-SEM results.

3.1. Deletion methods

Deletion methods, including listwise and pairwise deletion, remove observations with missing values from the analysis. These methods are computationally simple and often serve as default options in common statistical software (Wang et al., 2022). When data are MCAR with less than 5 % missing values, deletion methods can yield unbiased estimates of means and variances (Little & Rubin, 2019). Under these conditions, the simplicity and speed of deletion may be appealing in situations involving smaller datasets or low proportions of missing data.

However, even modest levels of missingness (5 %–20 %) can substantially reduce sample size and statistical power, particularly in smaller samples in many PLS-SEM research (Santos et al., 2019; Silva & Zárate, 2014). Moreover, if data are MAR or NMAR, deletion methods risk amplifying biases and distorting model estimates. Thus, while deletion methods may be appropriate for MCAR conditions, they are generally inadequate when missingness is systematically related to observed or unobserved variables.

3.2. EM imputation

In the context of missing values, imputation methods have been shown to be a superior approach when compared with deletion-based methods (Little & Rubin, 2019). This is due to the fact that imputation methods replace missing values with estimated ones, thereby preserving all observations and thus retaining sample size. Under MAR conditions, imputation often proves superior to deletion approaches because it exploits existing information to yield more reliable estimates (Little &

Rubin, 2019). One widely used technique is the EM algorithm, which iteratively refines estimates of missing data and corresponding model parameters until convergence (Wang et al., 2022; Enders, 2023). By leveraging maximum likelihood estimation, EM effectively manages MAR scenarios, maintaining inter-variable relationships and generally offering improved parameter estimation compared to deletion-based methods (Wang et al., 2022). It also adapts to various model structures, making it suitable for PLS-SEM environments. However, EM is not without its limitations. It has been observed to converge slowly when large portions of the data are missing, and it is somewhat sensitive to initial parameter assumptions (Jeong et al., 2021; Goretzko et al., 2020). While EM can be applied to NMAR data, it does not inherently adjust for biases arising from non-random missingness, necessitating additional corrections when data are NMAR. The following section provides an overview of the application of EM:

Step 1 is the expectation step (E step), where the algorithm estimates the missing values based on the current parameter estimates under the assumption that these parameters are known.

Step 2 is the maximization step (M step), during which those parameters are updated by maximizing the likelihood function to better fit both the observed data and the imputed values obtained from the E step.

Step 3 repeats the E and M steps in an iterative cycle until the changes in estimated parameters fall below a predefined threshold, indicating convergence. Once the EM algorithm converges, the completed dataset can be used to specify and evaluate a PLS-SEM model, benefiting from the preserved inter-variable relationships that often lead to more reliable estimates than deletion-based methods. The data sources, based on the corporate reputation model used by Hair et al. (2023, 2024), along with the R code and results for this procedure, are provided in the Appendix A. This model includes several constructs, such as *competence*, *likability*, *customer satisfaction*, and *customer loyalty*, each measured through multiple indicators.

3.3. Fully conditional specification and MICE

The fully conditional specification (FCS) method, often implemented via MICE, is a multiple imputation approach that draws on relationships among all variables in the dataset to generate plausible values for missing observations (Schafer & Graham, 2002; Grund et al., 2025). By creating multiple imputed datasets, FCS captures the uncertainty around the true values of missing points (Zheng et al., 2018). It is important to note that each dataset will differ slightly because of the randomness introduced in the imputation process (Grund et al., 2025), making MICE/FCS particularly suitable for data that are MAR.

The following section provides an overview of the application of FCS. The details of the R code for each step, as well as the final results, can be found in the Appendix A.

Step 1: The data were initially imported from an Excel file, with missing values systematically introduced based on predefined probabilities associated with factors such as customer satisfaction levels and service provider categories. To deal with missing data, multiple imputation was conducted using the *mice* package in R, employing a FCS approach with predictive mean matching. This method generated several complete datasets, each representing a plausible scenario for the missing values and capturing the uncertainty inherent in the imputation process.

Step 2: Subsequently, the PLS-SEM model was estimated for each imputed dataset. This involved defining the relevant constructs for corporate reputation and specifying the structural relationships between these variables. By analyzing each imputed dataset independently, estimates for path coefficients, reliability metrics, and fit indices were obtained, ensuring that the variability in these estimates reflected differences in the imputed values rather than inconsistencies in the modeling process.

Step 3: The results of the individual analyses were then aggregated to derive combined estimates, including average path coefficients,

standard errors, and confidence intervals. This aggregation provided a comprehensive view of the model parameters by accounting for the variability across multiple imputations. Additionally, the variability in path estimates and fit measures was examined to assess the potential presence of NMAR mechanisms. It is worth noting that significant variability or sensitivity to certain assumptions indicates that the missingness may be NMAR, highlighting the necessity of employing additional strategies, such as weighting or selection models, to reduce bias.

Aggregation or pooling of results is essential in FCS study, especially for handling missing data and variability across datasets (Jeong et al., 2021). It ensures robust and generalizable findings by averaging metrics such as outer loadings, reliability coefficients, and path coefficients across multiple datasets or iterations. This approach reduces the impact of outliers and enhances the consistency and validity of the model.

Using Rubin's rules (Little & Rubin, 2019) for pooling in PLS-SEM with multiple imputations and bootstrapping accounts for both within- and between-imputation variance, providing accurate intervals and reliable statistical inferences. Additionally, averaging predictive accuracy metrics (e.g., RMSE, MAE) across models offers a comprehensive view of the model's performance under various conditions. Pooling ensures that conclusions are stable and applicable across different scenarios, rather than being dependent on a single dataset. A detailed explanation and the calculation results are provided in the Appendix A.

3.4. Strategies for addressing NMAR data

The presence of NMAR data poses a particular challenge to PLS-SEM, given that the probability of missingness is contingent on unobserved values. This characteristic renders conventional imputation or deletion strategies susceptible to bias (Fiero et al., 2016). In circumstances where data are NMAR, the employment of advanced methodologies becomes imperative. Such methodologies encompass selection models, sensitivity analyses, and weighting strategies (Little & Rubin, 2019). Selection models, for instance, estimate both the data-generation and missingness processes concurrently, while weighting approaches such as inverse probability weighting (IPW) correct for selection bias by assigning greater influence to cases that are underrepresented (Cuddeback et al., 2004; Heckman, 1979; Seaman & White, 2013). These methods adjust the analysis rather than simply filling in missing values, thus preserving the validity of PLS-SEM estimates and interpretations.

Building on the limitations of simple imputation for NMAR data, this study proposes an integrated approach that combines EM-based imputation with a subsequent weighting procedure. EM first produces a complete dataset, allowing all observations to remain in the analysis, while the weighting step then accounts for non-random missingness. In line with recent advances highlighting the utility of EM for complex causal predictive models (Wang et al., 2022; Hair et al., 2023), this approach capitalizes on EM's robustness to MAR-like conditions and the capacity of weighting to correct NMAR-induced bias.

A critical component of the weighting procedure is modeling the likelihood of missing data. For instance, indicators of the competence construct in the corporate reputation model used by Hair et al. (2023, 2024)—*comp_1*, *comp_2*, and *comp_3*—can be selected as benchmark variables that drive missingness, based on the hypothesis that participants with lower assessments of the company's competence may be less inclined to disclose additional information. To capture this, a logistic regression is specified, where the probability of missingness is the dependent variable, and *comp_1*, *comp_2*, and *comp_3* serve as independent variables:

$$\log\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = \beta_0 + \beta_1 \cdot \text{comp}_1 + \beta_2 \cdot \text{comp}_2 + \beta_3 \cdot \text{comp}_3 + \epsilon \quad (\text{Equation 1}).$$

Where $P(Y = 1)$ is the probability of an observation being missing; β_0 , β_1 , β_2 , and β_3 are the model parameters to be estimated from the data; and $\log\left(\frac{P(Y=1)}{1-P(Y=1)}\right)$ is the logit (log-odds), a transformation

commonly used in logistic regression to model probability linearly.

Once the logistic regression has been fitted, missingness probabilities are predicted for each observation, and weights are calculated accordingly—often as the inverse of these probabilities or as a function of the minimum for *comp_1*, *comp_2*, and *comp_3*. For instance, if lower values of *comp_1*, *comp_2*, or *comp_3* correspond to higher missingness, the weight can be computed as:

$$\text{weight} = 1/(\min(\text{comp}_1, \text{comp}_2, \text{comp}_3) + c) \text{ (Equation 2).}$$

Where $\min(\text{comp}_1, \text{comp}_2, \text{comp}_3)$ denotes the minimum value among *comp_1*, *comp_2* and *comp_3*. *c* is a small constant that ensures the denominator never becomes zero.

The final step in the procedure entails conducting a weighted analysis of the imputed dataset—referred to here as weighted PLS-SEM (WPLS-SEM)—where cases are assigned weights that reflect their respective likelihood of being missing. This procedure helps mitigate

distortions due to selection and measurement bias, granting a more balanced representation of the data (Becker & Ismail, 2016; Cheah et al., 2021). By integrating EM-based imputation with weighting adjustments, researchers retain the benefits of a complete dataset while systematically correcting for the mechanisms that cause non-random missingness.

Unlike traditional two-step approaches, the EM-weighting method can operate in an iterative manner. After the initial EM imputation, the logistic regression for missingness is re-estimated using the newly imputed data, and updated weights are generated. Where appropriate, these recalibrated weights can then be fed back into a refined EM procedure, creating a feedback loop that more accurately captures the partial correlation between missingness and the underlying constructs. By iteratively refining both the imputation process and the missingness model, researchers can tailor the weighting to match the nuanced relationships within subsets of data — an important consideration when

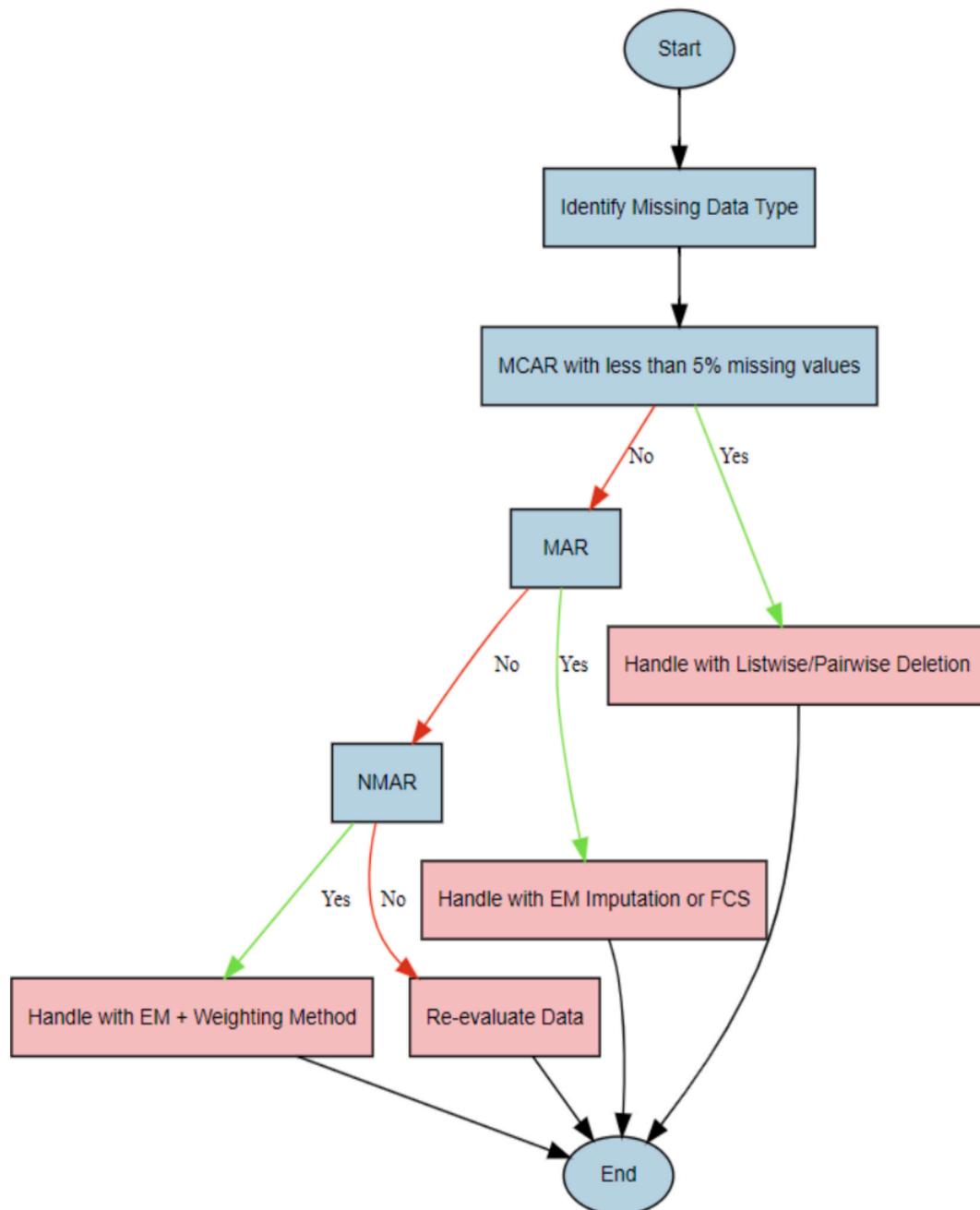


Fig. 1. Decision Framework for Missing Data Handling in Statistical Analysis.

indicators vary in how they influence or reflect missingness.

In multi-group analyses or when comparing different market segments or treatment groups, missingness mechanisms may differ between subpopulations. Modeling separate logistic regressions (or incorporating interaction terms in a single logistic model) for each group allows researchers to derive group-specific weights. This approach clarifies whether missingness in one segment is driven by different competence thresholds (e.g., $comp_1 < 3$) compared to another, thus providing a more nuanced and context-sensitive understanding of NMAR processes.

By preserving variance through EM and correcting bias through weighting, the integrated method reduces the likelihood of both over- and underestimating the relationships between constructs. This dual protection is particularly valuable for predictive models, where robust parameter estimates and reliable inferences guide strategic decision-making. Consequently, the iterative EM-weighting framework goes beyond simple post-hoc adjustments to address NMAR data, providing a dynamic solution that strengthens the validity of PLS-SEM analyses in multifaceted research settings. Fig. 1 outlines a step-by-step process for identifying the missing data mechanism (MCAR, MAR, NMAR) and selecting appropriate handling strategies, including imputation and weighting methods, based on diagnostic tests and analysis.

4. Simulation study on missing data thresholds

4.1. Purpose of the simulation

This simulation study examines the impact of varying levels of missing data on commonly used imputation methods in PLS-SEM,

$$prob(missing) = \frac{1}{1 + \exp[-(\beta_0 + \beta_1 \cdot cusa + \beta_2 \cdot serviceprovider + \beta_3 \cdot (cusa \cdot serviceprovider))]} \quad (3)$$

extending the analysis to include the NIPALS algorithm as a benchmark. By systematically varying the proportion and mechanism of missing values, we aim to identify the thresholds at which specific techniques produce unreliable or biased results, addressing concerns about method performance under increasing missingness. The study also assesses robustness by correlating latent variable scores with those from the complete dataset, as suggested by reviewers, to evaluate method interchangeability. The findings are intended to guide researchers in selecting appropriate missing data strategies, particularly as missingness severity increases, offering practical insights for business research applications.

4.2. Simulation design

We utilized a validated corporate reputation model (Hair et al., 2023, 2024), previously employed in Section 3, with an original dataset assumed to be complete and meeting PLS-SEM assumptions for data distribution and measurement properties. Three missing data mechanisms were introduced: MCAR (missing completely at random), MAR (missing at random), and NMAR (not missing at random). Missingness was categorized into five levels: 5 % (very low), 10 % (low), 25 % (moderate), 30 % (substantial), and 50 % (high), reflecting conditions commonly encountered in empirical research. These thresholds allow us to evaluate method performance across diverse scenarios.

To assess robustness, we computed the Pearson correlation between latent variable scores (*COMP*, *LIKE*, *CUSA*, and *CUSL*) from processed datasets and those from the complete dataset, averaging them to obtain Mean_Correlation. A Mean_Correlation above 0.9 suggests high robustness and potential interchangeability of methods, as per reviewer recommendations. This metric complements traditional indicators,

providing a comprehensive evaluation of method performance.

4.3. Simulation procedure

To evaluate the performance of missing data handling methods under varying mechanisms and missingness levels, we implemented a structured simulation procedure using the corporate reputation model described in Section 4.2. The following steps outline the process:

Step 1: Baseline analysis

We began by fitting the PLS-SEM model to the complete dataset to obtain baseline estimates of path coefficients, loadings, reliability metrics, and model fit indices. These baseline results serve as the “true” values against which the performance of different imputation methods is compared.

Step 2: Introducing missing data

For each of the missingness mechanisms (MCAR, MAR, and NMAR) and each of the percentage thresholds, the data were systematically removed according to specific criteria:

MCAR: Observations were removed entirely at random, independent of any other observed or unobserved variables. This approach simulates scenarios where missingness occurs purely by chance, such as accidental data loss.

MAR: Missingness was generated using a logistic regression-based approach to create a more realistic pattern of missing data. Instead of applying fixed probabilities based on a single variable, such as customer satisfaction (*cusa*), we incorporated multiple predictors and their interactions to determine the likelihood of missingness. The probability of missingness was modeled as follows:

This method allows the likelihood of missingness to decrease with higher satisfaction levels and to vary across different service provider categories. The inclusion of the interaction term between satisfaction and service provider makes the missingness mechanism more nuanced and reflective of real-world conditions.

NMAR: The simulation hypothesizes that respondents with lower competence assessments are more likely to withhold their scores. Specifically, *comp_1*, *comp_2*, and *comp_3* were identified as benchmark variables, with a threshold value (e.g., 2) established to determine missingness. If any of these scores fell below the threshold, the corresponding competence score was more likely to be missing. This design ensures that the missingness of each competence score is directly tied to its value rather than being influenced by other observed variables. Furthermore, the likelihood of missing data was explicitly modeled to increase as the scores decreased below the threshold, reinforcing the NMAR framework.

Step 3: Applying imputation methods

For each incomplete dataset generated under the different mechanisms and thresholds, we applied the following imputation methods:

- **Deletion:** Listwise deletion, removing incomplete cases, as a baseline to assess bias and data loss.
- **EM Imputation:** Used the EM algorithm (via *mice* with *norm* method, $m = 1$) to estimate missing values under the MAR assumption.
- **MICE/FCS:** Generated multiple imputations ($m = 5$, *pmm* method) to capture uncertainty, averaging results across imputations.
- **EM-Weighting:** Combined EM imputation with inverse probability weighting (IPW) to address NMAR conditions, computing weights

based on competence scores and applying them in PLS-SEM (weights implementation pending validation in SmartPLS 4, Ringle et al., 2023).

- **NIPALS:** Applied directly via SEMinR's built-in algorithm (Hair et al., 2021), handling missing data iteratively without imputation, generating latent scores for comparison.

Step 4: PLS-SEM analysis

After imputation, we re-estimate the PLS-SEM analysis on each completed dataset. We then recorded the parameter estimates, model fit indices, path coefficients, R^2 values, and reliability measures. This step allowed us to assess how well each imputation method recovered the true model parameters.

Step 5: Evaluation metrics

We assessed method performance using six indicators compared against the baseline:

- **Mean_Bias:** Average absolute deviation of path coefficients from baseline values; lower values indicate better accuracy.
- **MSE:** Mean squared error of path coefficients; lower values indicate less estimation variance.
- **Coverage:** Percentage of bootstrap confidence intervals (2.5 %–97.5 %) containing baseline path coefficients; higher values indicate greater reliability.
- **Convergence_Rate:** Percentage of runs reaching a stable solution; higher values indicate computational stability.
- **Time_Per_Run:** Average computational time per run (seconds); lower values indicate higher efficiency.
- **Mean_Correlation:** Average Pearson correlation of latent variable scores with baseline scores; values closer to 1 (especially > 0.9) indicate higher robustness.

Notably, the EM-Weighting's Time_Per_Run was not recorded due to separate weighting computation, with subsequent analyses performed in SmartPLS 4, which supports WPLS-SEM but does not track time consistently with other methods.

Step 6: Repetition and aggregation

Each condition (mechanism and threshold combination) was repeated 100 times to ensure estimate stability. Results were aggregated across iterations to compute average Mean_Bias, MSE, Coverage, Convergence_Rate, Time_Per_Run, and Mean_Correlation, enabling robust conclusions about method performance.

Table 3
Comparison of Methods Across Mechanisms and Levels.

Mechanism	Missingness Level	Deletion	EM	FCS	EM-Weighting	NIPALS
MCAR	5 %	+	++	++	++	++
	10 %	+	++	++	++	++
	25 %	–	+	++	++	+
	30 %	–	+	+	++	+
	50 %	–	–	–	+	–
MAR	5 %	+	++	++	++	++
	10 %	+	++	++	++	++
	25 %	–	+	++	++	+
	30 %	–	+	+	++	+
	50 %	–	–	–	+	–
NMAR	5 %	+	+	+	++	+
	10 %	–	+	+	++	+
	25 %	–	–	–	+	–
	30 %	–	–	–	+	–
	50 %	–	–	–	–	–

Legend: ++ (Best: low bias, high correlation, high coverage), + (Good: moderate bias, acceptable correlation), – (Poor: high bias, low correlation, low coverage).

4.4. Results

The simulation reveals distinct performance patterns for Deletion, EM, FCS, EM-Weighting, and NIPALS across MCAR, MAR, and NMAR mechanisms at 5 %, 10 %, 25 %, 30 %, and 50 % missingness levels. Table 3 summarizes method performance using symbols (++, + for good, – for poor) based on Mean_Bias, Mean_Correlation, and Coverage, while Fig. 2 visualizes trends across five metrics. Complete numerical results are provided in Appendix B, Table B1.

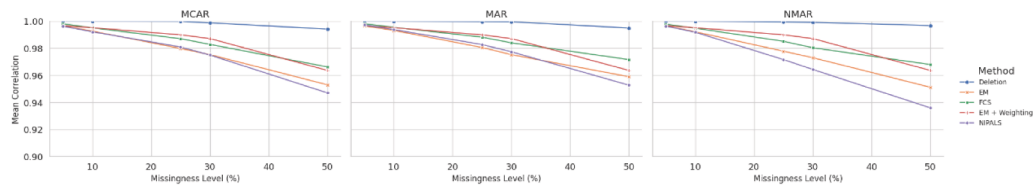
Across MCAR and MAR conditions, EM-Weighting and FCS consistently outperform others, earning ++ ratings at 5 %–30 % missingness due to high Mean_Correlation (> 0.9635) and low biases (< 0.0343). EM and NIPALS perform well at low to moderate missingness (5 %–25 %), rated ++ or +, but decline to – at 50 % in MCAR (Mean_Correlation 0.9529 and 0.9469) due to increased complexity. Deletion struggles above 10 %, rated at 25 %–50 % with high bias (up to 0.1396 in MCAR 50 %) and reduced Coverage (e.g., 84 % at MCAR 30 %), reflecting sample loss. In NMAR, EM-Weighting excels at 5 %–30 % (++ rating, Mean_Correlation up to 0.9966), followed by FCS, but all methods falter at 50 % (– rating), with NIPALS showing the lowest robustness (Mean_Correlation 0.9359, Coverage 96 %).

NIPALS matches EM and FCS at low missingness (5 %–25 %) in MCAR and MAR but underperforms in NMAR at higher levels, highlighting its limitations for non-random missingness. Fig. 2 illustrates these trends: Subplot 1 (Mean_Correlation) shows EM-Weighting and FCS's robustness, with NIPALS declining in NMAR; Subplot 2 (Mean_Bias) confirms EM-Weighting's minimal bias; Subplot 3 (Coverage) highlights Deletion's decline; Subplot 4 (MSE) shows EM-Weighting and FCS's low variance; and Subplot 5 (Time_Per_Run) reveals FCS's efficiency (0.66–1.05 s). All methods converged stably, so Convergence_Rate was not reported. EM-Weighting's computational time and FCS's Coverage require further validation due to incomplete data.

4.5. Discussion of simulation findings

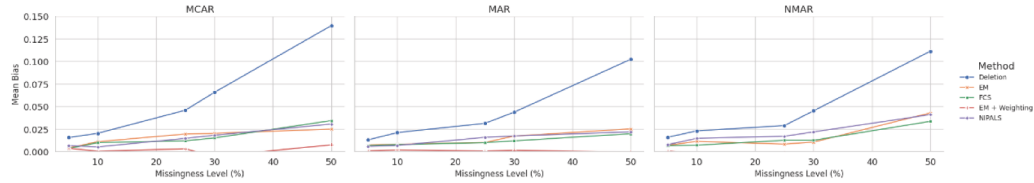
The simulation findings, based on 100 iterations per condition, highlight the critical need to align missing data management strategies with the underlying mechanism and extent of missingness, as outlined in Section 4.1. Table 3 provides a clear comparison of method performance across MCAR, MAR, and NMAR at 5 %, 10 %, 25 %, 30 %, and 50 % missingness levels, using symbolic ratings (++, +, –) based on Mean_Bias, Mean_Correlation, and Coverage. In MCAR scenarios, EM-Weighting and FCS consistently outperform others, earning ++ ratings at 5 %–30 %, due to their ability to maintain high robustness (Mean_Correlation > 0.9635) and low biases. EM and NIPALS perform comparably at 5 %–25 % (rated ++ or +), effectively leveraging observed data relationships, but decline to – at 50 % due to increased complexity, as their assumptions (e.g., MAR for EM) are less tenable with higher missingness. Deletion struggles above 10 % (rated – at 25 %–50 %), introducing significant bias and variance due to sample reduction, making it unsuitable for most practical applications.

Under MAR, EM-Weighting and FCS again lead, rated ++ up to 30 %, capitalizing on observed covariates (e.g., customer satisfaction, service provider) to maintain robust latent score correlations (Mean_Correlation > 0.9635). EM and NIPALS match their performance at 5 %–25 % (rated ++ or +), but at 50 % (rated –), NIPALS's iterative approach falters with complex missingness patterns, while EM struggles to fully account for systematic bias. Deletion's performance deteriorates above 10 % (rated at 25 %–50 %), as systematic missingness amplifies bias, rendering it inadequate for MAR scenarios. In NMAR, where missingness depends on unobserved factors (e.g., competence scores below a threshold), EM-Weighting excels at 5 %–30 % (rated ++), mitigating bias through inverse probability weighting. FCS performs well at 5 %–25 % (rated ++), but all methods falter at 50 % (rated –), with NIPALS showing the most pronounced limitations due to its inability to address unobserved drivers of missingness.



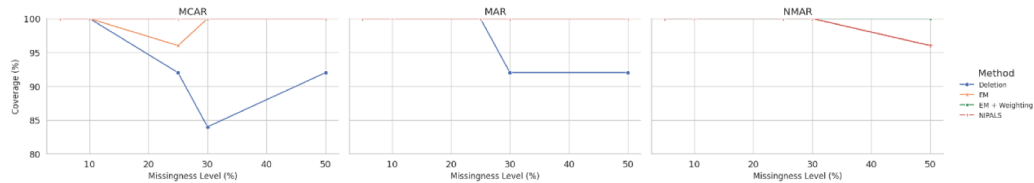
Subplot 1: Mean Correlation of Latent Variable Scores Across Missingness Mechanisms

Line plots showing the Mean_Correlation of latent variable scores for Deletion, EM, FCS, EM-Weighting, and NIPALS across MCAR, MAR, and NMAR mechanisms at 5%, 10%, 25%, 30%, and 50% missingness levels. EM-Weighting and FCS maintain high correlations (> 0.96), while NIPALS declines in NMAR at 50% (0.9359).



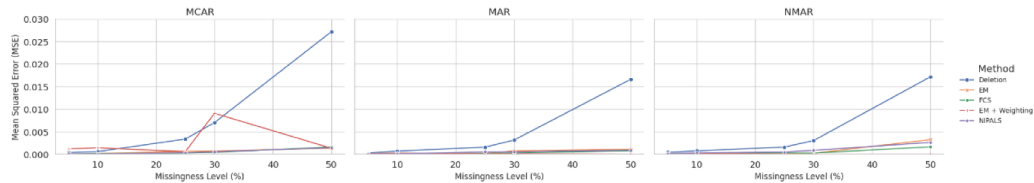
Subplot 2: Mean Bias Across Missingness Mechanisms

Line plots showing the Mean_Bias for Deletion, EM, FCS, EM-Weighting, and NIPALS across MCAR, MAR, and NMAR mechanisms at 5%, 10%, 25%, 30%, and 50% missingness levels. EM-Weighting exhibits minimal bias (< 0.0074), while Deletion's bias peaks at 0.1396 (MCAR 50%).



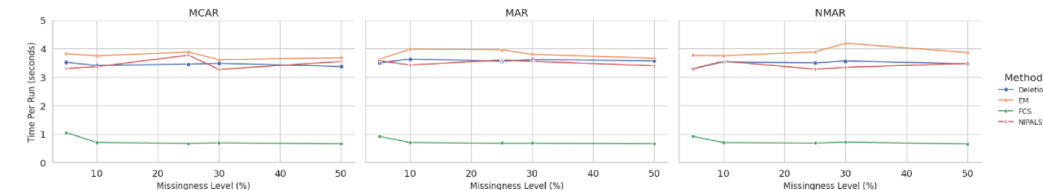
Subplot 3: Coverage Across Missingness Mechanisms

Line plots showing Coverage for Deletion, EM, EM-Weighting, and NIPALS across MCAR, MAR, and NMAR mechanisms at 5%, 10%, 25%, 30%, and 50% missingness levels (FCS excluded due to incomplete data). Deletion's Coverage drops to 84% (MCAR 30%), while EM-Weighting remains at 100%.



Subplot 4: Mean Squared Error (MSE) Across Missingness Mechanisms

Line plots showing the Mean Squared Error (MSE) for Deletion, EM, FCS, EM-Weighting, and NIPALS across MCAR, MAR, and NMAR mechanisms at 5%, 10%, 25%, 30%, and 50% missingness levels. EM-Weighting and FCS maintain low variance (< 0.0016), while Deletion peaks at 0.0271 (MCAR 50%).



Subplot 5: Time Per Run Across Missingness Mechanisms

Line plots showing the Time_Per_Run for Deletion, EM, FCS, and NIPALS across MCAR, MAR, and NMAR mechanisms at 5%, 10%, 25%, 30%, and 50% missingness levels (EM-Weighting excluded due to incomplete data). FCS is the most efficient (0.66–1.05 seconds), while EM and NIPALS range from 3.2–4.2 seconds.

Fig. 2. Performance Metrics by Missingness Level and Mechanism (Subplots 1–5).

Fig. 2 visualizes these trends: Subplot 1 (Mean_Correlation) confirms EM-Weighting and FCS's robustness across mechanisms, while Subplot 2 (Mean_Bias) highlights Deletion's increasing bias at higher missingness. Subplot 3 (Coverage) shows Deletion's decline, Subplot 4 (MSE) underscores EM-Weighting and FCS's low variance, and Subplot 5 (Time_Per_Run) reveals FCS's computational efficiency, making it practical for large datasets. These insights lead to three key recommendations for business researchers: (1) prefer imputation or weighting methods (e.g., EM, FCS, EM-Weighting) over Deletion above 10 % missingness to avoid significant bias and ensure robust path estimates; (2) match the method to the missingness mechanism—use EM, FCS, or NIPALS for MCAR/MAR up to 25 %, and EM-Weighting for NMAR scenarios to address unobserved factors; (3) consider advanced solutions like EM-Weighting for NMAR at 30 %–50 % missingness, where standard methods struggle. These guidelines support the study's aim to help researchers select appropriate strategies, ensuring reliable PLS-SEM analyses in business contexts where missing data is prevalent.

5. Discussion

Effective management of missing data is pivotal in PLS-SEM, where complex structural models and even modest missingness can skew results (Becker et al., 2023; Hwang et al., 2023; Sarstedt et al., 2023). This study underscores the need to align missing data strategies with the underlying mechanism—MCAR, MAR, or NMAR—to minimize bias and preserve model integrity. Our simulation (Section 4.4) reveals that EM-Weighting excels in NMAR at 5 %–30 %, maintaining high robustness (Mean_Correlation > 0.987) and low biases (< 0.0018), while FCS offers efficiency (0.66–1.05 s per run) and strong performance in MCAR and MAR (Mean_Correlation > 0.9635). NIPALS performs well at low missingness (5 %–25 %) in MCAR and MAR but struggles in NMAR at 50 %, highlighting its limitations for non-random missingness. Deletion becomes unreliable above 10 %, making it unsuitable for MAR and NMAR.

The choice of method depends on the percentage and nature of missingness. For small fractions (e.g., 5 %) under MCAR, simple approaches like Deletion or NIPALS may suffice, as they are unbiased and efficient (Table 4). Above 10 % missingness, particularly in MAR, methods like EM or FCS are necessary to leverage observed data relationships. For NMAR exceeding 30 %, EM-Weighting is essential to mitigate severe bias. These thresholds, detailed in Section 4.5, guide researchers in selecting context-appropriate strategies, ensuring the accuracy of PLS-SEM parameters despite data loss.

This study introduces a novel EM-Weighting approach for NMAR, addressing a critical gap in non-random missingness literature. Its iterative synergy—re-estimating missingness parameters and feeding refined weights back into the EM algorithm—captures subtle data relationships, enhancing reliability. The method's flexibility is able to support multi-group analyses, accommodating both linear and nonlinear relationships, and mitigates statistical errors, ensuring robust parameter estimates for predictive models in strategic decision-making.

Table 4 summarizes the pros and cons of each method, aiding

business researchers in method selection. For instance, EM-Weighting excels in NMAR but requires complex implementation, while NIPALS offers simplicity for low missingness scenarios. To further support method selection, Table 5 provides an aggregate performance analysis, averaging key metrics across all simulation conditions and assigning overall performance ratings (++ , + , -). EM-Weighting and FCS lead with ++ ratings, excelling in robustness (Mean_Correlation 0.9859–0.9873) and low biases (< 0.0133), while Deletion ranks lowest (-) due to high bias (0.0430) and variance (MSE 0.0044). Together, Tables 4 and 5 offer a comprehensive guide for researchers, combining qualitative insights with quantitative performance data to ensure robust PLS-SEM analyses in business contexts.

It is incumbent upon researchers to identify missing data patterns in a timely manner. In the event that the selected PLS algorithm (e.g., NIPALS) is capable of accommodating missing items, the utilization of these features may be preferable to deletion, particularly in complex models. However, it should be noted that each method has its limitations. EM can be slow to converge, FCS depends on correct imputation model specifications, and NMAR-oriented techniques like EM-Weighting are less supported by standard software. A systematic approach, encompassing the documentation of decisions from the classification of missingness mechanisms to the configuration of parameters, has been demonstrated to enhance the credibility, transparency, and reliability of PLS-SEM analyses.

5.1. Theoretical contributions

This study advances the theoretical understanding of missing data in PLS-SEM by addressing several gaps in the literature, contributing to the knowledge of how missing data impacts model reliability and method selection. First, it provides empirical evidence on the performance of missing data handling methods under NMAR conditions, where missingness depends on unobserved factors. Existing PLS-SEM literature often focuses on MCAR and MAR scenarios, with limited attention to NMAR due to its complexity (Wang et al., 2022; Amusa & Hossana, 2024). Our findings demonstrate that EM-Weighting effectively mitigates bias in NMAR scenarios up to 30 % missingness (Mean_Correlation > 0.987, biases < 0.0018), offering a theoretical foundation for applying weighting techniques in non-random missingness contexts. This extends theoretical discussions by providing a benchmark for NMAR handling, suggesting that advanced methods can partially address unobserved drivers of missingness, a critical area for future theoretical development.

Second, the study identifies and quantifies critical missingness thresholds that impact method reliability, refining theoretical frameworks for method selection in PLS-SEM. Although prior studies have recognized that method performance is contingent on the severity of missingness (Wang et al., 2022), these investigations have not systematically delineated explicit guidelines for determining a particular threshold. Our results show that Deletion becomes unreliable above 10 % missingness (bias up to 0.1396 at 50 %), while NIPALS struggles in NMAR at 50 % (Mean_Correlation 0.9359, Coverage 96 %). These

Table 4
Pros and Cons of Missing Data Handling Methods in PLS-SEM.

Method	Pros	Cons
Deletion	Simple, fast, unbiased for MCAR with low missingness (<5%)	Reduces sample size, high bias for MAR/NMAR, unsuitable for high missingness
EM	Preserves sample size, effective for MCAR/MAR, maintains relationships	Limited for NMAR, slow convergence at high missingness, sensitive to assumptions
FCS	Captures uncertainty, suitable for MCAR/MAR, flexible for complex models	Computationally intensive, less effective for NMAR, requires careful specification
EM-Weighting	Excels in NMAR, reduces bias, supports multi-group analyses	Complex implementation, computationally demanding, less effective at extreme NMAR
NIPALS	Built-in PLS-SEM algorithm, handles low MCAR/MAR missingness, generates latent scores for ANOVA/MANOVA/clustering	No imputed dataset, limited for NMAR, less effective at high missingness

Table 5
Aggregate Performance Analysis of Missing Data Handling Methods.

Method	Avg Mean_Correlation	Avg Mean_Bias	Avg MSE	Avg Coverage	Avg Time_Per_Run (s)	Overall Performance
Deletion	0.9985	0.0430	0.0044	96.8	3.5	–
EM	0.9788	0.0145	0.0005	99.7	3.8	+
FCS	0.9873	0.0133	0.0003	NA	0.7	++
EM-Weighting	0.9859	0.0004	0.0012	100	NA	++
NIPALS	0.9760	0.0159	0.0006	99.7	3.5	+

Notes:

- Averages are computed across all conditions (MCAR, MAR, NMAR; 5 %, 10 %, 25 %, 30 %, 50 % missingness). For each metric (Mean_Correlation, Mean_Bias, MSE, Coverage, Time_Per_Run), values were averaged over the 15 conditions (3 mechanisms × 5 levels) per method, excluding NA values (e.g., FCS's Coverage, EM-Weighting's Time_Per_Run).
- To compute the Overall Performance ratings, each metric was normalized to a 0–1 scale: for Mean_Correlation and Coverage, higher values were scaled closer to 1 (e.g., $\text{Mean_Correlation} = (\text{value} - \min) / (\max - \min)$); for Mean_Bias, MSE, and Time_Per_Run, lower values were scaled closer to 1 (e.g., $\text{Mean_Bias} = 1 - (\text{value} - \min) / (\max - \min)$). A composite score was then calculated by averaging the normalized scores across all metrics, with equal weights. Methods were assigned ratings based on this score: ++ (top 40th percentile), + (middle 40th–60th percentile), – (bottom 40th percentile).
- Coverage excludes FCS, and Time_Per_Run excludes EM-Weighting due to incomplete data.

thresholds offer a theoretical foundation for understanding methodological limitations, thereby strengthening frameworks such as those proposed by Hair et al. (2021) with evidence-based guidance on when to transition from basic to advanced analytical techniques. This is especially relevant in business research, where datasets often contain varying levels of missing data due to factors such as survey nonresponse or data collection challenges. Consequently, there is a critical need for robust theoretical models to guide the selection of appropriate and effective methods.

Third, by integrating latent score correlations (Mean_Correlation) as a robustness metric, this study introduces a new theoretical perspective on method performance evaluation in PLS-SEM. Traditional metrics like Mean_Bias and MSE focus on parameter accuracy, but Mean_Correlation assesses the stability of latent variable scores (*COMP*, *LIKE*, *CUSA*, and *CUSL*), which are central to PLS-SEM's predictive capabilities. While robustness in PLS-SEM has been discussed in terms of parameter recovery (Becker et al., 2023), the use of latent score correlations as a complementary metric is novel. Values above 0.9 for EM-Weighting and FCS across most conditions suggest potential method interchangeability, aligning with theoretical discussions on robustness (Jeong et al., 2021). This metric responds to recent calls for multidimensional evaluation in structural equation modeling, providing a framework for future research to explore alternative robustness measures. For instance, it could inform theoretical models that prioritize predictive accuracy over parameter precision in business applications, where understanding latent constructs like customer satisfaction is often more critical than exact path coefficients.

Finally, the study's findings have broader theoretical implications for PLS-SEM frameworks, particularly in the context of model complexity and sample size constraints. PLS-SEM is often chosen in business research for its ability to handle complex models with predictive analysis capabilities (Hair et al., 2024), but missing data can exacerbate issues like bias and variance. Our results suggest that advanced methods like EM-Weighting and FCS can maintain model integrity even at higher missingness levels (up to 30 %), challenging the assumption that PLS-SEM's robustness diminishes significantly with missing data (Wang et al., 2022). This contributes to theoretical debates on the applicability of PLS-SEM in real-world settings, where missingness is common, by demonstrating that method choice can mitigate these challenges, thereby expanding the theoretical boundaries of PLS-SEM's utility.

5.2. Methodological contributions

This research expands the methodological toolkit for PLS-SEM by systematically comparing Deletion, EM, FCS, EM-Weighting, and NIPALS under MCAR, MAR, and NMAR conditions. It provides a

comprehensive framework—encompassing practical guidelines, empirical examples, and decision-making criteria—that enables researchers to select context-appropriate strategies. The study's detailed illustration within a corporate reputation model demonstrates how these methods can be operationalized, bridging the gap between theoretical principles and empirical implementation. For instance, in business contexts like customer satisfaction studies, where missing data often arises from survey non-response, this framework guides researchers in choosing methods that preserve sample size and ensure robust path estimates. The inclusion of NIPALS as a benchmark method further enriches the toolkit by offering a built-in algorithmic solution for low missingness scenarios, which can be particularly useful in exploratory analyses where latent scores are needed for further statistical tests (e.g., ANOVA, clustering).

The novel EM-Weighting approach enhances methodological flexibility, particularly for complex PLS-SEM applications. Its iterative synergy—re-estimating missingness parameters and feeding refined weights back into the EM algorithm—captures subtle data relationships, improving reliability over traditional static methods. This dynamic feedback loop allows the method to adapt to the unique characteristics of each dataset, accommodating both linear and nonlinear relationships. For example, in a multi-group analysis of customer satisfaction, demographic groups (e.g., age, income level) may exhibit distinct missingness patterns driven by group-specific factors (e.g., higher non-response among younger respondents due to survey fatigue). EM-Weighting models these variations by applying group-specific weights or interaction terms (e.g., satisfaction levels and income), ensuring a nuanced understanding of missingness mechanisms. This adaptability makes it a valuable tool for business researchers dealing with heterogeneous datasets, where understanding segment-specific effects is critical for strategic decision-making.

Moreover, the integrated strategy mitigates common statistical challenges, such as Type I and Type II errors, by preserving variance through EM and correcting bias via weighting. This is particularly critical for predictive models in business contexts, where inaccurate imputations could lead to flawed predictions and suboptimal outcomes. For instance, in a corporate reputation study predicting customer loyalty (*CUSL*), inaccurate handling of missing competence scores (*COMP*) could overestimate or underestimate relationships, leading to misguided marketing strategies. EM-Weighting's ability to maintain robust parameter estimates ensures more reliable predictions, supporting evidence-based decision-making. The method's support for multi-group analyses further extends its utility, allowing researchers to explore how missingness patterns vary across segments (e.g., by region or customer type), a common requirement in business research applications.

The study also provides methodological advancements through its evaluation framework, which combines traditional metrics (Mean_Bias, MSE, Coverage) with the novel Mean_Correlation metric. This

multidimensional approach allows researchers to assess method performance from both parameter accuracy and predictive robustness perspectives (Wang et al., 2022; Amusa & Hossana, 2024). The framework’s practical guidelines—such as preferring imputation over Deletion above 10 % missingness—provide a decision-making tool that can be applied across diverse PLS-SEM applications, from marketing to organizational behavior studies. By operationalizing these methods within a real-world corporate reputation model, the study ensures that methodological contributions are not only theoretically sound but also practically actionable, enhancing the credibility, transparency, and replicability of PLS-SEM analyses in the presence of missing data.

5.3. Limitations and future studies

This study offers valuable insights but does not cover all data types, models, or missing data patterns. Future research could apply these missing value methods to more complex PLS-SEM models, such as multi-group analyses, longitudinal data, etc. Additional simulations might identify critical missingness thresholds for different methods, helping researchers choose techniques based on data characteristics. Emerging machine learning and Bayesian approaches (Sarstedt et al., 2023) could also improve NMAR data handling in PLS-SEM, enhancing precision and adaptability for modern empirical research.

6. Conclusion

This study provides critical insights into missing data handling in PLS-SEM, evaluating Deletion, EM, FCS, EM-Weighting, and NIPALS across MCAR, MAR, and NMAR mechanisms at 5 %, 10 %, 25 %, 30 %, and 50 % missingness levels. Key findings highlight that EM-Weighting and FCS consistently outperform other methods in NMAR and MAR scenarios, maintaining high robustness (Mean_Correlation > 0.9635) and low biases (< 0.0343) up to 30 % missingness. Deletion becomes

unreliable above 10 % missingness, exhibiting significant bias (up to 0.1396 at 50 %) and reduced Coverage (84 %–92 %), making it unsuitable for MAR and NMAR. NIPALS, while effective for low missingness (5 %–25 %) in MCAR and MAR, struggles in NMAR at 50 % (Mean_Correlation 0.9359), underscoring its limitations for non-random missingness. FCS offers computational efficiency (0.66–1.05 s per run), making it practical for large business datasets.

These findings are pivotal for PLS-SEM research, particularly in business contexts where missing data is prevalent due to survey non-response or incomplete records. The identified thresholds—10 % for Deletion’s unreliability and 50 % for NIPALS’s limitations—provide actionable guidelines for researchers to select methods based on missingness levels and mechanisms. The superiority of EM-Weighting in NMAR scenarios advances the application of advanced techniques in non-random missingness contexts, ensuring more accurate path estimates and preserving sample size. By integrating latent score correlations (Mean_Correlation) as a robustness metric, this study offers a new lens for evaluating method performance, enhancing the reliability of PLS-SEM models in predictive and exploratory analyses.

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CRedit authorship contribution statement

Yide Liu: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Wynne W. Chin:** Supervision, Methodology, Conceptualization. **Jun-Hwa Cheah:** Writing – review & editing, Methodology, Conceptualization. **Joseph F. Hair:** Writing – review & editing. **Chan Lyu:** Writing – review & editing, Funding acquisition, Data curation.

Appendix

Appendix A.: PLS-SEM analysis results for different data handling methods

The data sources, based on the corporate reputation model used by Hair et al. (2023, 2024), along with the R code and results for this procedure, are provided at: <https://github.com/yide109/missing.git>.

This section study presents the results of PLS-SEM for various methods of handling missing data, including EM, FSC, and EM with weighting adjustments. Each method was applied to the same dataset to evaluate its impact on the analysis outcomes and to determine which provided the most robust insights into the data under study.

For comparative purposes, this Appendix also includes PLS-SEM results based on the original data without any treatment of missing data. From an overall perspective of the results, most indicators (loadings, reliability and validity indicators, discriminant validity, and path coefficients) under the four treatment methods are quite close to each other. This suggests that despite the different process of missing value imputation, the core measurement model and conclusions regarding most structural paths have not undergone drastic changes. For example, most external loadings are between 0.77 and 0.92, and the direction of the path coefficients remains largely consistent.

1. Outcomes of PLS-SEM analysis using EM imputed data

Table A1-1
Outer loadings.

Construct	Indicator	Loading
comp	comp_1: [The company] is a top competitor in its market.	0.853
	comp_2: As far as I know, [the company] is recognized worldwide.	0.794
	comp_3: I believe that [the company] performs at a premium level.	0.815
cusa	cusa: I have a positive attitude towards [the company].	1.000
cusi	cusi_1: I would recommend [company] to friends and relatives.	0.838
	cusi_2: If I had to choose again, I would choose [company] as my mobile phone services provider.	0.921
	cusi_3: I will remain a customer of [company] in the future.	0.847

(continued on next page)

Table A1-1 (continued)

Construct	Indicator	Loading
like	like_1: [The company] is a company that I can better identify with than other companies.	0.880
	like_2: [The company] is a company that I would regret more not having if it no longer existed than I would other companies.	0.870
	like_3: I regard [the company] as a likeable company.	0.843

Table A1-2

Construct reliability and validity.

Construct	Cronbach's alpha	Composite reliability (rho_a)	Composite reliability (rho_c)	Average variance extracted (AVE)
comp	0.770	0.824	0.861	0.674
cusl	0.837	0.845	0.903	0.756
like	0.831	0.836	0.899	0.747

Table A1-3

Discriminant validity – Heterotrait–monotrait ratio (HTMT).

Construct	comp	cusa	cusl	like
comp				
cusa	0.466			
cusl	0.536	0.749		
like	0.795	0.579	0.739	

Table A1-4

Path coefficients.

Relationship	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	95 % CI Lower	95 % CI Upper
comp → cusa	0.153	0.154	0.067	0.043	0.262
comp → cusl	0.005	0.007	0.055	−0.08	0.099
cusa → cusl	0.498	0.497	0.042	0.427	0.566
like → cusa	0.430	0.430	0.061	0.328	0.530
like → cusl	0.352	0.351	0.057	0.256	0.445

Table A1-5

PLSpredict result.

Indicator	Q ² predict	PLS-SEM_RMSE	PLS-SEM_MAE	LM_RMSE	LM_MAE
cusa	0.282	0.997	0.778	0.993	0.780
cusl_1	0.311	1.259	0.948	1.266	0.939
cusl_2	0.325	1.435	1.073	1.450	1.094
cusl_3	0.214	1.477	1.090	1.496	1.115

2. Outcomes of PLS-SEM analysis using FSC imputed data

The use of pooled results is indispensable when conducting research using FSC (Jeong et al., 2021), particularly when addressing missing data or variability among multiple datasets. Pooling results provides a robust and generalized understanding of the relationships and metrics within the model being studied, ensuring the stability and robustness of the findings across various datasets or model iterations. Averaging results across multiple iterations can mitigate the impact of outliers or anomalies present in any single dataset, leading to more accurate and reliable conclusions.

Pooling results from different datasets or models also enhances the generalizability of findings, making the conclusions applicable to a broader context. To obtain pooled results for outer loadings, for instance, one would calculate the average loading for each construct across multiple datasets or model iterations. This method emphasizes the consistency of indicators in measuring their respective constructs. Similarly, averaging reliability metrics such as Cronbach's Alpha, composite reliability (rhoC), average variance extracted (AVE), and composite reliability (rhoA) across various models can help reliably assess the internal consistency and validity of the constructs.

Additionally, by averaging pooled heterotrait–monotrait ratio (HTMT) values across multiple analyses, researchers can evaluate how distinct a construct is from other constructs, ensuring that constructs do not measure the same phenomenon. Calculating the average path coefficients across multiple model estimations can further confirm the strength and direction of relationships between constructs, ensuring that these relationships hold across different conditions or imputations.

When managing multiple imputations in PLS-SEM analysis, especially when incorporating bootstrapping, using methods like Rubin's rules is critical. This approach ensures that estimates from each imputed dataset are accurately combined, reflecting the uncertainty across multiple imputations and providing robust statistical inferences.

The first step in pooling results involves calculating the pooled estimates for the original estimates and the bootstrap means across all imputations.

The pooled standard deviation is then determined by calculating the square root of the sum of the average within-imputation variance and the between-imputation variance, as specified by Rubin's rules. This comprehensive approach not only accounts for the variance within each set of imputations but also considers the variability between constructed datasets, offering a more thorough error estimate. Next, confidence intervals are calculated using the overall pooled standard deviation and the pooled means. These intervals are pivotal in assessing the range within which the true values of the estimates are expected to lie, taking into account the uncertainty introduced by the missing data.

For predictive accuracy metrics such as root mean square error (RMSE) and mean absolute error (MAE), these values are averaged across different predictive models or datasets to provide a holistic view of the model's predictive performance under various conditions. The aggregation of findings through pooled results is crucial to ensure that the conclusions drawn from the PLS-SEM model are not based merely on a single dataset or limited conditions but are robust across a variety of scenarios.

Table A2-1

Pooled outer loadings.

Construct	Indicator	Loading
comp	comp_1: [The company] is a top competitor in its market.	0.857
	comp_2: As far as I know, [the company] is recognized worldwide.	0.796
	comp_3: I believe that [the company] performs at a premium level.	0.813
cusa	cusa: I have a positive attitude towards [the company].	1.000
cusi	cusi_1: I would recommend [company] to friends and relatives.	0.834
	cusi_2: If I had to choose again, I would choose [company] as my mobile phone services provider.	0.919
	cusi_3: I will remain a customer of [company] in the future.	0.845
like	like_1: [The company] is a company that I can better identify with than other companies.	0.879
	like_2: [The company] is a company that I would regret more not having if it no longer existed than I would other companies.	0.870
	like_3: I regard [the company] as a likeable company.	0.843

Table A2-2

Pooled construct reliability and validity.

Construct	Cronbach's alpha	Composite reliability (rho_a)	Composite reliability (rho_c)	Average variance extracted (AVE)
comp	0.772	0.863	0.676	0.832
cusi	0.834	0.900	0.751	0.841
like	0.831	0.899	0.747	0.836

Table A2-3

Pooled discriminant validity – Heterotrait–monotrait ratio (HTMT).

Construct	comp	cusa	cusi	like
comp				
cusa	0.463			
cusi	0.528	0.754		
like	0.788	0.580	0.737	

Table A2-4

Pooled path coefficients.

Relationship	Pooled Original Estimate	Pooled Bootstrap Mean	Pooled SD	95 % CI Lower	95 % CI Upper
comp → cusa	0.154	0.158	0.069	0.023	0.293
comp → cusi	0.079	0.082	0.068	−0.050	0.215
cusa → cusi	0.505	0.504	0.042	0.422	0.587
like → cusa	0.431	0.429	0.062	0.307	0.552
like → cusi	0.565	0.562	0.062	0.441	0.684

Table A2-5
Pooled PLSpredict results.

Indicator	PLS-SEM_RMSE	PLS-SEM_MAE	LM_RMSE	LM_MAE
cusa	0.997	0.778	0.846	0.671
cusl_1	1.203	0.888	1.216	0.893
cusl_2	1.245	0.925	1.258	0.934
cusl_3	1.312	0.957	1.328	0.964

3. Outcomes of PLS-SEM analysis using weighted EM imputed data

Table: A3-1. Outer loadings

Construct	Indicator	Loading
comp	comp_1: [The company] is a top competitor in its market.	0.848
	comp_2: As far as I know, [the company] is recognized worldwide.	0.793
	comp_3: I believe that [the company] performs at a premium level.	0.776
cusa	cusa: I have a positive attitude towards [the company].	1.000
cusl	cusl_1: I would recommend [company] to friends and relatives.	0.850
	cusl_2: If I had to choose again, I would choose [company] as my mobile phone services provider.	0.929
	cusl_3: I will remain a customer of [company] in the future.	0.825
like	like_1: [The company] is a company that I can better identify with than other companies.	0.894
	like_2: [The company] is a company that I would regret more not having if it no longer existed than I would other companies.	0.875
	like_3: I regard [the company] as a likeable company.	0.843

Table A3-2

Construct reliability and validity.

Construct	Cronbach's alpha	Composite reliability (rho_a)	Composite reliability (rho_c)	Average variance extracted (AVE)
comp	0.747	0.809	0.848	0.650
cusl	0.836	0.845	0.902	0.755
like	0.841	0.852	0.904	0.758

Table A3-3

Discriminant validity – Heterotrait–monotrait ratio (HTMT).

Construct	comp	cusa	cusl	like
comp				
cusa	0.474			
cusl	0.546	0.762		
like	0.808	0.571	0.758	

Table A3-4

Path coefficients.

Relationship	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	95 % CI Lower	95 % CI Upper
comp → cusa	0.165	0.167	0.078	0.039	0.294
comp → cusl	−0.017	−0.011	0.059	−0.107	0.086
cusa → cusl	0.501	0.499	0.048	0.418	0.576
like → cusa	0.419	0.417	0.066	0.307	0.524
like → cusl	0.388	0.384	0.058	0.289	0.479

Table A3-5
PLSpredict result.

Indicator	Q ² predict	PLS-SEM_RMSE	PLS-SEM_MAE	LM_RMSE	LM_MAE
cusa	0.283	0.995	0.776	0.992	0.776
cusl_1	0.311	1.262	0.950	1.268	0.943
cusl_2	0.323	1.439	1.078	1.452	1.099
cusl_3	0.205	1.484	1.099	1.503	1.124

4. Outcomes of PLS-SEM analysis using Original data

Table A4-1
Outer loadings.

Construct	Indicator	Loading
comp	comp_1: [The company] is a top competitor in its market.	0.858
	comp_2: As far as I know, [the company] is recognized worldwide.	0.798
	comp_3: I believe that [the company] performs at a premium level.	0.818
cusa	cusa: I have a positive attitude towards [the company].	1.000
cusl	cusl_1: I would recommend [company] to friends and relatives.	0.833
	cusl_2: If I had to choose again, I would choose [company] as my mobile phone services provider.	0.917
	cusl_3: I will remain a customer of [company] in the future.	0.843
like	like_1: [The company] is a company that I can better identify with than other companies.	0.879
	like_2: [The company] is a company that I would regret more not having if it no longer existed than I would other companies.	0.870
	like_3: I regard [the company] as a likeable company.	0.843

Table A4-2
Construct reliability and validity.

	Cronbach's alpha	Composite reliability (rho_a)	Composite reliability (rho_c)	Average variance extracted (AVE)
comp	0.776	0.832	0.865	0.681
cusl	0.831	0.839	0.899	0.748
like	0.831	0.836	0.899	0.747

Table A4-3
Discriminant validity – Heterotrait–monotrait ratio (HTMT).

Construct	comp	cusa	cusl	like
comp				
cusa	0.465			
cusl	0.532	0.755		
like	0.780	0.577	0.737	

Table A4-4
Path coefficients.

Relationship	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	95 % CI Lower	95 % CI Upper
comp → cusa	0.162	0.162	0.065	0.054	0.268
comp → cusl	0.009	0.012	0.054	−0.076	0.103
cusa → cusl	0.504	0.503	0.042	0.433	0.572
like → cusa	0.424	0.425	0.059	0.327	0.522
like → cusl	0.342	0.341	0.056	0.248	0.433

Table A4-5
PLSpredict result.

Indicator	Q ² predict	PLS-SEM_RMSE	PLS-SEM_MAE	LM_RMSE	LM_MAE
cusa	0.282	0.996	0.777	0.993	0.778
cusl_1	0.307	1.258	0.945	1.265	0.940
cusl_2	0.320	1.434	1.076	1.448	1.097
cusl_3	0.204	1.474	1.089	1.492	1.111

Appendix B

Table B1

Complete simulation results

Missingness_Mechanism	Missingness_Level	Method	Mean_Bias	MSE	Coverage	Convergence_Rate	Time_Per_Run	Mean_Correlation
MCAR	5	Deletion	0.015477	0.00039	100	100	3.518419	0.999942
MCAR	5	EM	0.003873	0.000025	100	100	3.818301	0.996464
MCAR	5	FCS	0.003289	0.000020	NA	100	1.048673	0.997982
MCAR	5	EM + Weighting	0.0036	0.001188	100	NA	NA	0.996614
MCAR	5	NIPALS	0.006593	0.000086	100	100	3.304109	0.996207
MCAR	10	Deletion	0.0202	0.000583	100	100	3.405705	0.99989
MCAR	10	EM	0.010906	0.000186	100	100	3.746163	0.992619
MCAR	10	FCS	0.009825	0.00015	NA	100	0.703316	0.995014
MCAR	10	EM + Weighting	0.0004	0.001416	100	NA	NA	0.995148
MCAR	10	NIPALS	0.00516	0.000071	100	100	3.36386	0.992116
MCAR	25	Deletion	0.045792	0.003357	92	100	3.454737	0.999751
MCAR	25	EM	0.019424	0.000634	96	100	3.878581	0.979733
MCAR	25	FCS	0.011886	0.000286	NA	100	0.675031	0.986985
MCAR	25	EM + Weighting	0.003	0.000578	100	NA	NA	0.989865
MCAR	25	NIPALS	0.014582	0.00034	100	100	3.768667	0.980884
MCAR	30	Deletion	0.065641	0.00701	84	100	3.484416	0.99872
MCAR	30	EM	0.020184	0.000696	100	100	3.611087	0.97524
MCAR	30	FCS	0.015035	0.000431	NA	100	0.691995	0.982774
MCAR	30	EM + Weighting	-0.0052	0.009068	100	NA	NA	0.987085
MCAR	30	NIPALS	0.018044	0.00054	100	100	3.269805	0.974919
MCAR	50	Deletion	0.139568	0.027146	92	100	3.370099	0.994132
MCAR	50	EM	0.024779	0.001341	100	100	3.679884	0.952908
MCAR	50	FCS	0.034282	0.001594	NA	100	0.660811	0.966155
MCAR	50	EM + Weighting	0.0074	0.001328	100	NA	NA	0.963529
MCAR	50	NIPALS	0.03047	0.001446	100	100	3.547123	0.946943
MAR	5	Deletion	0.012904	0.000293	100	100	3.499271	0.999959
MAR	5	EM	0.007292	0.000107	100	100	3.631174	0.996461
MAR	5	FCS	0.006788	0.000086	NA	100	0.918104	0.99798
MAR	5	EM + Weighting	0.0008	0.00011	100	NA	NA	0.996614
MAR	5	NIPALS	0.005714	0.000048	100	100	3.572141	0.997147
MAR	10	Deletion	0.020995	0.000697	100	100	3.632193	0.999859
MAR	10	EM	0.008013	0.000127	100	100	3.979916	0.993115
MAR	10	FCS	0.007676	0.000149	NA	100	0.700618	0.995464
MAR	10	EM + Weighting	0.0018	0.000271	100	NA	NA	0.995148
MAR	10	NIPALS	0.006742	0.000108	100	100	3.422545	0.99375
MAR	25	Deletion	0.031197	0.001562	100	100	3.555577	0.999591
MAR	25	EM	0.01008	0.000182	100	100	3.959796	0.980485
MAR	25	FCS	0.009819	0.000171	NA	100	0.678044	0.988113
MAR	25	EM + Weighting	0.0006	0.000078	100	NA	NA	0.989865
MAR	25	NIPALS	0.015713	0.000529	100	100	3.597892	0.982661
MAR	30	Deletion	0.04361	0.003133	92	100	3.609885	0.999506
MAR	30	EM	0.017128	0.000746	100	100	3.798824	0.975221
MAR	30	FCS	0.011831	0.000284	NA	100	0.678453	0.983926
MAR	30	EM + Weighting	0.0014	0.000118	100	NA	NA	0.987085
MAR	30	NIPALS	0.017265	0.000522	100	100	3.550209	0.977343
MAR	50	Deletion	0.102316	0.016623	92	100	3.569752	0.994917
MAR	50	EM	0.025126	0.001104	100	100	3.66325	0.958892
MAR	50	FCS	0.01974	0.000782	NA	100	0.66346	0.971573
MAR	50	EM + Weighting	-0.0004	0.00002	100	NA	NA	0.963529
MAR	50	NIPALS	0.02181	0.000897	100	100	3.397353	0.952655
NMAR	5	Deletion	0.015826	0.000428	100	100	3.288341	0.999959
NMAR	5	EM	0.006903	0.000128	100	100	3.762982	0.996553
NMAR	5	FCS	0.006618	0.000094	NA	100	0.922115	0.997682
NMAR	5	EM + Weighting	0.0008	0.00012	100	NA	NA	0.996614
NMAR	5	NIPALS	0.007616	0.000107	100	100	3.290092	0.996304
NMAR	10	Deletion	0.022888	0.000741	100	100	3.534995	0.999875
NMAR	10	EM	0.011245	0.0002	100	100	3.753048	0.99211
NMAR	10	FCS	0.007067	0.000089	NA	100	0.69912	0.994782
NMAR	10	EM + Weighting	-0.0018	0.000207	100	NA	NA	0.995148
NMAR	10	NIPALS	0.014653	0.000318	100	100	3.558499	0.991763
NMAR	25	Deletion	0.028738	0.001586	100	100	3.497032	0.999401
NMAR	25	EM	0.00825	0.000113	100	100	3.881497	0.977755
NMAR	25	FCS	0.012516	0.000309	NA	100	0.684133	0.985065
NMAR	25	EM + Weighting	-0.0008	0.000043	100	NA	NA	0.989865
NMAR	25	NIPALS	0.016934	0.0005	100	100	3.279084	0.971599
NMAR	30	Deletion	0.045181	0.003019	100	100	3.562567	0.999071
NMAR	30	EM	0.010443	0.000231	100	100	4.187939	0.97295
NMAR	30	FCS	0.012461	0.000271	NA	100	0.713815	0.980308
NMAR	30	EM + Weighting	-0.0014	0.000014	100	NA	NA	0.987085

(continued on next page)

Table B1 (continued)

Missingness_Mechanism	Missingness_Level	Method	Mean_Bias	MSE	Coverage	Convergence_Rate	Time_Per_Run	Mean_Correlation
NMAR	30	NIPALS	0.021717	0.00086	100	100	3.340322	0.96425
NMAR	50	Deletion	0.111261	0.017157	100	100	3.469022	0.996727
NMAR	50	EM	0.042826	0.003217	96	100	3.862619	0.951102
NMAR	50	FCS	0.033458	0.001629	NA	100	0.655771	0.967892
NMAR	50	EM + Weighting	−0.0018	0.000012	100	NA	NA	0.963529
NMAR	50	NIPALS	0.040942	0.002615	96	100	3.478971	0.935873

Notes:

- **Simulation Setup:** Results are based on 50 iterations per condition, using the corporate reputation model (Hair et al., 2023, 2024) with PLS-SEM on a dataset assumed to be complete before introducing missingness at 5 %, 10 %, 25 %, 30 %, and 50 % levels.
- **Metrics:**
 - *Mean_Bias:* Average absolute deviation of path coefficients from baseline values (complete dataset); lower values indicate better accuracy.
 - *MSE:* Mean squared error of path coefficients; lower values indicate less estimation variance.
 - *Coverage:* Percentage of bootstrap confidence intervals (2.5 %–97.5 %) containing baseline path coefficients; higher values indicate greater reliability.
 - *Time_Per_Run:* Average computational time per run (seconds); lower values indicate higher efficiency.
 - *Mean_Correlation:* Average Pearson correlation of latent variable scores (COMP, LIKE, CUSA, CUSL) with baseline scores; values closer to 1 (especially > 0.9) indicate higher robustness.
- **Missing Values:**
 - Coverage for FCS is marked as 'NA' due to incomplete implementation of confidence interval calculations across multiple imputations (m = 5).
 - Time_Per_Run for EM-Weighting is marked as 'NA' due to incomplete computational data; ongoing efforts aim to resolve this in future analyses.
- **Interpretation:** EM-Weighting and FCS consistently show high Mean_Correlation (> 0.96) and low biases, particularly in NMAR, while Deletion exhibits significant bias at higher missingness (e.g., 0.1396 at MCAR 50 %). Coverage and Time_Per_Run for some methods require further validation due to missing data.

Data availability

The data and code supporting the findings of this study are available from the corresponding author upon reasonable request. Additionally, the replication materials, including the dataset and R code for the analyses, are openly accessible at <https://github.com/yide109/missing>.

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